

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA14 COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO3: Analyze syntax-directed translation method using synthesized and inherited attributes to generate a Parser Tree. (BL4)

Assessment Tool Weightage: 10%

Duration :60 Mins

On Class

QUESTIONS: 50 Total Marks:50

Annexure A

SCENARIO BASED TASK

Code of Conduct:

I Vishal B (192524485) certify that this submission is my original Work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Vishal B

CO3 - ASSESSMENT

TOOL - 2

Name : B.vishal

Reg no: 192524485

Course Name : Compiler Design

Course code : CSA-1408

1. Syntax - Directed Definition (SDD):

$$\alpha. S \rightarrow E_n \{ S.val = E.val \}$$

$$\alpha. E \rightarrow E_1 + T \{ E.val = E_1.val + T.val \}$$

$$\alpha. F \rightarrow E_1 - T \{ E.val = E_1.val - T.val \}$$

$$\alpha. E \rightarrow T \{ E.val = T.val \}$$

$$\alpha. T \rightarrow T_1 * F \{ T.val = T_1.val * F.val \}$$

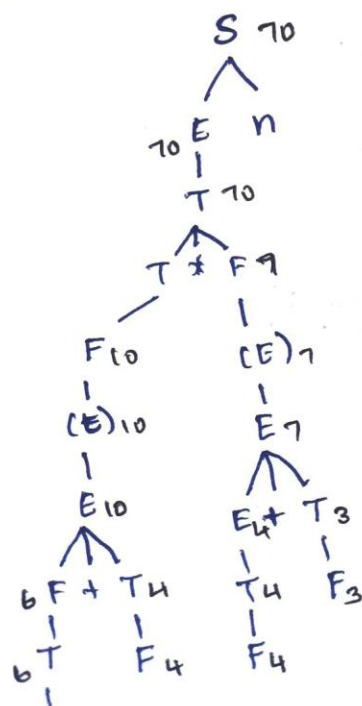
$$\alpha. T \rightarrow T_1 / F \{ T.val = F.val \}$$

$$\alpha. T \rightarrow F \{ T.val = F.val \}$$

$$\alpha. F \rightarrow (E) \{ F.val = E.val \}$$

$$\alpha. F \rightarrow \text{digit} \{ F.val = \text{digit.lexical} \}$$

Annotated Parse tree for $(6+4) * (4+3)$:



(val = 70)

4. Left subtree: $(b+4)$

$$F \rightarrow \text{digit}(6) \Rightarrow F.\text{val} = 6$$

$$T \rightarrow F \Rightarrow T.\text{val} = 6$$

$$E \rightarrow T \Rightarrow E.\text{val} = 6$$

$$F \rightarrow \text{digit}(4) \Rightarrow F.\text{val} = 4$$

$$T \rightarrow F \Rightarrow T.\text{val} = 4$$

$$E \rightarrow E(6) + T(4) = E.\text{val}(10)$$

$$F \rightarrow (E(10)) \Rightarrow F.\text{val} = 10$$

$$T \rightarrow F \Rightarrow T.\text{val} = 10$$

4. Right subtree: $(4+3)$

$$F \rightarrow \text{digit}(4) \Rightarrow F.\text{val} = 4$$

$$T \rightarrow F \Rightarrow T.\text{val} = 4$$

$$E \rightarrow T \Rightarrow E.\text{val} = 4$$

$$F \rightarrow \text{digit}(3) \Rightarrow F.\text{val} = 3$$

$$T \rightarrow F \Rightarrow T.\text{val} = 3$$

$$E \rightarrow E(4) + T(3) = F.\text{val} = 7$$

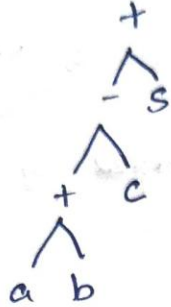
$$F \rightarrow (E(7)) \Rightarrow F.\text{val} = 7$$

Final computed value : 70

2. Syntax tree for:

$a * b - c + 8$

4. considering standard operator precedence ($* \rightarrow - = +$) and left to right associativity.



3. Syntax - Directed Definition (SDD):

1. $E \rightarrow E + T \{ E \cdot \text{Node} = \text{new node}(' + ', E \cdot \text{Node}, T \cdot \text{Node}) \}$

2. $E \rightarrow E - T \{ E \cdot \text{Node} = \text{new node}(' - ', E \cdot \text{Node}, T \cdot \text{Node}) \}$

3. $E \rightarrow T \{ E \cdot \text{Node} = T \cdot \text{Node} \}$

4. $T \rightarrow T * F \{ T \cdot \text{Node} = \text{new node}(' * ', T \cdot \text{Node}, F \cdot \text{Node}) \}$

5. $T \rightarrow F \{ T \cdot \text{Node} = F \cdot \text{Node} \}$

6. $F \rightarrow id \{ F \cdot \text{Node} = \text{new leaf}(id, id, \text{entry}) \}$

7. $F \rightarrow \text{num} \{ F \cdot \text{Node} = \text{new leaf}(\text{num}, \text{num.val}) \}$

4. Attribute Evaluation sequence:

1.) Leaf 1 = new leaf(id, a)

2.) Leaf 2 = new leaf(id, b)

3.) Node 1 = new node('*', leaf 1, leaf 2)

4.) Leaf 3 = new leaf(id, c)

5.) Node 2 = new node('-', node 1, leaf 3)

6.) Leaf 4 = new leaf(num, 8)

7.) Node 3 = new node('+', node 2, leaf 4)

3. L-Attributed Definition conditions:

• An SDD is L-attributed if each inherited attribute of a symbol on the right hand side of a production $A \rightarrow Y_1 Y_2 \dots Y_n$ only on.

• The inherited attributes of the head non-terminal A .

• The attributes synthesized or inherited of symbol $Y_1, Y_2, \dots, Y_{n-1}$ located to the left of Y_n .

• Analysis of Rule sets:

• Rule 1 ($A \rightarrow PQ$):

$$\alpha. P.i = A.i \rightarrow 2;$$

Inherited attribute of P depends on $A.i$ (head attribute)
 $\hookrightarrow$ valid.

$$\alpha. Q.i = P.i \rightarrow A.i;$$

Inherited attribute of Q depends on $P.i$ (symbol to its left) and $A.i$ (head attribute) valid.
 $\hookrightarrow$

$$\alpha. A.s = P.s \rightarrow Q.s;$$

Synthesized attribute of A depends on children $P.s$ and $Q.s$
 $\hookrightarrow$ valid

• Rule 2 ($A \rightarrow xy$):

$$\alpha. X.i = A.i \rightarrow Y.s;$$

Inherited attribute of X depends on $Y.s$ which is a symbol to its

right $\rightarrow$ violation.

$$\alpha. Y.i = X.s \rightarrow A.i$$

Inherited attribute of Y depends on $X.s$ (symbol to its left) and $A.i$
 $\hookrightarrow$ valid.

$\therefore$ There by,

• satisfying $L \rightarrow$ attribute definition : Rule 1

• violates $L \rightarrow$ attribute definition : Rule 2 [due to the right to left dependency]

4. $S \rightarrow$ Attributed syntax - Directed definition

- * $S \rightarrow E$; $\{ S.code = E.code ; \text{print}(S.code) ; \}$
- * $E \rightarrow E_1 + T$; $\{ E.code = E_1.code \parallel T.code \parallel "+" ; \}$
- * $E \rightarrow E_1 - T$; $\{ E.code = E_1.code \parallel T.code \parallel "-" ; \}$
- * $E \rightarrow T$; $\{ E.code = T.code ; \}$
- * $T \rightarrow T_1 * F$; $\{ T.code = T_1.code \parallel F.code \parallel "*" ; \}$
- * $T \rightarrow T_1 / F$; $\{ T.code = T_1.code \parallel F.code \parallel "/" ; \}$
- * $T \rightarrow F$; $\{ T.code = F.code ; \}$
- * $F \rightarrow (E)$; $\{ F.code = E.code ; \}$
- * $F \rightarrow \text{digit}$; $\{ F.code = \text{digit.lexical} ; \}$

* LR parsing and Attribute evaluation Tree (2+5*6):

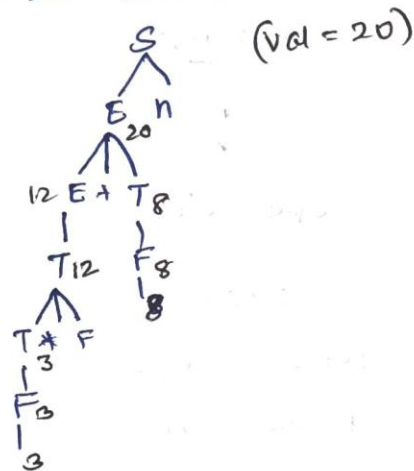
step	stack	Input	operation	computed attribute
1	\$	2+5*6; \$	shift 2	-
2	\$ digit +2	+5*6; \$	Reduce F $\rightarrow$ digit	F.code = '2'
3	\$F	+5*6; \$	Reduce T $\rightarrow$ F	T.code = '2'
4	\$T	+5*6; \$	Reduce E $\rightarrow$ T	E.code = '2'
5	\$E	+5*6; \$	shift +	-
6	\$E+	+5*6; \$	shift 5	E.code = '5'
7	\$E+digit	*6; \$	Reduce F $\rightarrow$ digit	T.code = '5'
8	\$E+F	*6; \$	Reduce T $\rightarrow$ F	-
9	\$E+T	*6; \$	shift *	-
10	\$E+T*	6; \$	shift 6	F.code = '6'
11	\$E+T*di(6)	; \$	Reduce F $\rightarrow$ digit	T.code = "56"
12	\$E+T*F	; \$	Reduce E $\rightarrow$ E+T	E.code = "256"
13	\$E+T	; \$	shift ;	-
14	\$E	; \$	shift	S.code = "256"
15	\$E	\$	$S \rightarrow E$;	Intermediate postfix
16	\$S	\$	Accept	Note generated.

Postfix intermediate code = 256*+

5. Syntax - Directed Definition (SDD):

- * $S \rightarrow E n \{ S.val = E.val \}$
- * $E \rightarrow E + T \{ E.val = E1.val + T.val \}$
- * $E \rightarrow T \{ E.val = T.val \}$
- * $T \rightarrow T * F \{ T.val = T.val * F.val \}$
- * $T \rightarrow F \{ T.val = F.val \}$
- * $F \rightarrow (E) \{ F.val = E.val \}$
- * $F \rightarrow \text{digit} \{ F.val = \text{digit.lexical} \}$

* Annotated Parse tree for: $3 * 4 + 8 n$



* Top-Down Evaluation sequence of Final output:

- 1.) Parse $F \rightarrow \text{digit}(3) \Rightarrow F.val = 3$
- 2.) Parse $T \rightarrow F \Rightarrow T.val = 3$
- 3.) Parse $F \rightarrow \text{digit}(4) \Rightarrow F.val = 4$
- 4.) Compute $T \rightarrow T(3) * F(4) \Rightarrow T.val = 3 * 4 = 12$
- 5.) Compute $E \rightarrow T(12) \Rightarrow E.val = 12$
- 6.) Parse $F \rightarrow \text{digit}(8) \Rightarrow F.val = 8$
- 7.) Compute $T \rightarrow F(8) \Rightarrow T.val = 8$
- 8.) Compute $E \rightarrow E(12) + T(8) \Rightarrow E.val = 12 + 8 = 20$
- 9.) Compute $S \rightarrow E(20) n \Rightarrow S.val = 20$

Final computed value = 20